CS 3750 In-Class Discussion 1

You want to build a new social networking site. The first thing to do is to implement user account management. Feel free to have a look at Gmail or Facebook to get clues. You may need to consider all the aspects of user account management system carefully.

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1. How will the user account management look like?

The user account management will include a registration page, login page, profile page, settings page, home page, an admin interface, and a coder interface.

2. What are the use cases? What are some user stories?

Some of the use cases here could be user registration, user login, user logout, password reset, edit profile, delete account, search all profiles and/or content, restrict accounts, grant access to accounts, edit software

As a [type of user], I want [an action], so that [some reason]
Some of the user stories could be:

- As an end-user, I want to create an account using my email and password, so I can save my actions taken on the site.
- As an end-user, I want to login with my email and password, so I can access my account.
- As an end-user, I want to logout, so I can leave my account saved and so others can't access it.
- As an end-user, I want to reset my password if I forgot it.

- As an end-user, I want to edit my profile, so I can change my description, photo, contact information, login information, or other basic information.
- As an end-user, I want to delete my account, so I can remove my information from the site and the database.
- As an administrator, I want to restrict accounts, so I can eliminate spam and hackers from the platform.
- As an administrator, I want to grant access to an account, so I can better help the users of the software.
- As a coder, I want to edit the software, so I can ensure the system runs smoothly.

3. What are the different components of user account management?

- Frontend: registration, login, profile management, logout
- Backend: carries out registration, authentication, login, changing information
- Database: stores user data, login details, settings
- Security: encryption, captcha
- Admin Interface: manages accounts
- Coder Interface: manages software

4. What are functional requirements?

- User registration with authentication
- Password reset with email or challenge questions
- User based access (i.e. a user and a moderator) and their permissions
- Profile update
- Profile deletion
- Account management
- Software management
- Usability (user-friendly interface)

5. What are some non-functional requirements?

- Ability to scale
- Platform stays running
- Security
- Fast processing speed
- Easy to use and learn (navigation)

- Compatibility with multiple browsers and devices

6. How will the UI look like?

- Simple design to allow quick adoption
- Person Icon for profile page and account information
- House icon for home page
- 3 dashes for settings menu
- All user settings will be at the top of the page
- Interaction will happen directly above and below the desired content.

7. What technologies will you use? If you must build a fully functional end-to-end user management system.

- Frontend: C# with windows or HTML, CSS with a website hoster like Bluehost
- Backend: Django(Python) or Java with APIs
- Database: SQL
- Security: Web Application firewalls

8. What are possible errors that may occur during user account management? How will you handle it?

- Invalid Login: Error message stating the login information is incorrect. Warning message saying the account will be locked after 3 failed login attempts. Offer password reset.
- Lockout After Failed Attempts: could temporarily prevent login attempts for a particular username/email for a period of time
- Duplicate Account: Message saying Email already in use. Offer password reset.
- Invalid email address: check and warn if the user is trying to enter an email address that doesn't have a valid format (*text@text.text*)
- Changing the password to a currently used or previously used password.
- Servers are down: Display Error message popup giving user relevant details regarding the shutdown.

9. Anything else you want to share?

The sketching and planning of this project should be extensive to try to cover the vast majority if not all of the first draft's use cases, technologies in use, and a flexible work distribution.

10. If you have 4 team members, how many hours will it take to build such a system?

Without ever having made a project to this scale it feels difficult to project a timeline, especially with 4 team members. Here's my rough estimate:

- Preparation: 20 hours
- Frontend: 40 hours
- Backend: 40 hours
- Database: 30 hours
- Security: 35 hours
- Testing & Debugging: 24 hours

Approximately 189 working hours between the 4 team members